

PHSDSC717P – Mathematical Methods IV Lab

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Part I

Assignments

1 Complex Roots

- (a) Using the two-dimensional Newton-Raphson method, find the cube roots of a complex number by forming two simultaneous equations from the real and imaginary parts.
- (b) Calculate the roots by de Moivre's formula and compare the results.
- (c) Find complex roots of an arbitrary algebraic equation.

HINTS AND RELEVANT PYTHON FUNCTIONS:

- For obtaining the modulus and argument of a complex number, you may use `cmath.polar`.
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2 Improper Integrals

Consider the integrals, $I_c(R) = \oint_C f(z)dz$ and $I_r(R) = \int_{-R}^R f(x)dx$, C being a closed contour going from $-R$ to R along the real axis and coming back along a semi-circular contour of radius R centered at the origin and on the upper half of the complex plane. For each of the functions, (i) $f(z) = \frac{1}{z^2 + 1}$ (ii) $f(z) = \frac{\cos 3z}{z^4 + 81}$
(iii) $f(z) = \frac{e^{5iz}}{z^6 + 64}$,

(a) evaluate $\int_{-\infty}^{\infty} f(x)dx \equiv \lim_{R \rightarrow \infty} I_r(R)$,

(b) calculate the contribution from the semi-circular arc, as $R \rightarrow \infty$, and,

(c) calculate $\lim_{R \rightarrow \infty} I_c(R)$.

HINTS AND RELEVANT PYTHON FUNCTIONS:

- Use Simpson's $\frac{1}{3}$ rule for the integrals by writing your own code.
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Part II

Theory

1 Two-dimensional Newton-Raphson Method

Ref: Press et. al., *Numerical Recipes in Fortran*

Problem: Find a root of $\vec{F}(\vec{x}) = 0$ using a guess value $\vec{x}_0$.

Expanding $\vec{F}(\vec{x})$ in a Taylor series to linear order, we have,

$$\vec{F}(\vec{x} + \delta\vec{x}) = \vec{F}(\vec{x}) + J(\vec{x})\delta\vec{x}, \quad (1)$$

where $J(\vec{x})$ is the Jacobian matrix, i.e., $J_{ij} = \frac{\partial F_i}{\partial x_j}$. If $\vec{x} + \delta\vec{x}$ is a root, then the left hand side of the above equation must vanish. Then, we have,

$$J\delta\vec{x} = -\vec{F} \Rightarrow \delta\vec{x} = -J^{-1}\vec{F}. \quad (2)$$

Thus, the algorithm for finding the root is to change $\vec{x}$ following this rule until convergence is achieved.

$$\vec{x}_{n+1} = \vec{x}_n - J^{-1}\vec{F}. \quad (3)$$

For a two-dimensional problem, J is a 2×2 matrix. Thus, if,

$$J(\vec{x}) = \begin{pmatrix} a(\vec{x}) & b(\vec{x}) \\ c(\vec{x}) & d(\vec{x}) \end{pmatrix}, \text{ then, } J^{-1}(\vec{x}) = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}. \quad (4)$$

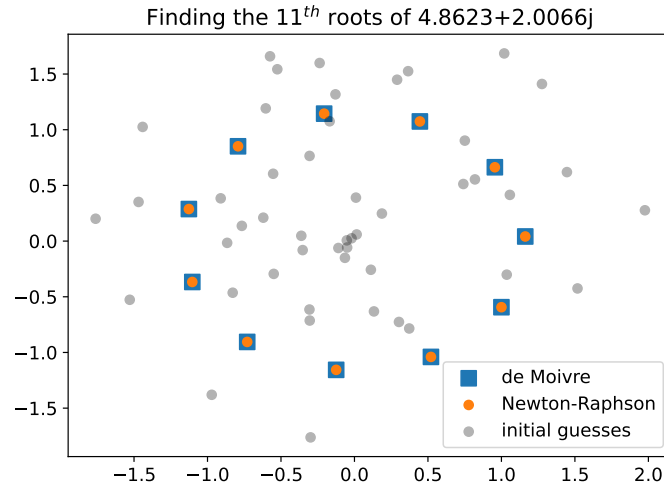


Figure 1: Sample calculation of roots with two-dimensional Newton-Raphson method and comparison with exact answers.